Dhritabrata Mitra

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SKILLS

Languages: Java, JavaScript (ES6+), TypeScript, Python, SQL
Frontend: React, Next.js, HTML, CSS, Tailwind
Backend: Spring Boot, Node.js, REST APIs, GraphQL
Data & Messaging: PostgreSQL, MongoDB, Redis, Kafka, RabbitMQ, Elasticsearch
Cloud & DevOps: AWS (EC2, EKS, S3), Docker, Kubernetes, Terraform, Jenkins, Git, Linux
Testing & Security: JUnit, Jest, React Testing Library, Playwright, Cypress, Selenium, JWT, OAuth

WORK EXPERIENCE

Cognizant Sept 2020 – Jun 2023

Software Engineer

Project: Estee Lauder

- Engineered a distributed e-commerce system using Next.js, TypeScript, Java, Spring Boot, Selenium, MongoDB, and PostgreSQL, serving 500K+ monthly users, contributing to $15B+ in annual revenue.
- Addressed peak-traffic order notification bottlenecks by refactoring legacy synchronous Java workflows into an event-driven pipeline, using Kafka for high-volume order events and RabbitMQ for reliable notification delivery, boosting throughput by 30%.
- Modernized e-commerce product search to resolve latency and scalability issues during high traffic, using Redis caching and Elasticsearch, reducing search latency by 40% and backend load by 30%.
- Strengthened API security by implementing a JWT-secured rate-limiting microservice with Spring Security and Redis, enforcing per-user limits and reducing security incidents by 40%.
- Refactored legacy frontend components to modern React patterns and optimized REST and GraphQL APIs, eliminating UI inconsistencies and slow data loads, reducing page errors by 30%, and improving overall page responsiveness.
- Resolved inefficient product browsing on category pages by enhancing filtering, sorting, and pagination, resulting in a 10% increase in session duration and 20% higher customer engagement.

Project: Capital One

- Stabilized production deployments by implementing secure CI/CD pipelines using Jenkins on Linux, AWS CodeDeploy, and IAM, reducing manual deployment errors by 30%.
- Cut release cycles by 50% by deploying Spring Boot, Node.js, and React services on AWS EC2 and EKS behind Application Load Balancers, standardizing deployments with Docker, Kubernetes, and Terraform (IaC).
- Improved frontend reliability and performance by applying TDD with unit and end-to-end testing (Jest, React Testing Library, Cypress, Playwright) and optimizing bundles through lazy loading and Webpack code splitting.
- Participated in production on-call rotations, diagnosing and resolving high-priority incidents using AWS CloudWatch and distributed logs.
- Mentored 5 interns, led code reviews, and introduced shared coding standards that significantly reduced post-release defects across multiple deployments.

Software Engineer Intern Jan 2020 – Apr 2020

- Resolved 10+ critical bugs and developed 5+ new features, improving application performance and reducing crash rates by 35% within 4 months.

EDUCATION

M.S. in Computer Science, Michigan Technological University, GPA: 3.85/4.0 Aug 2023 – Apr 2025
B.Tech in Computer Science, University of Engineering & Management, GPA: 3.39/4.0 Jul 2016 – May 2020

TECHNICAL PROJECTS

Meet AI | Demo | Live Site | GitHub

- Built a SaaS platform with OpenAI LLM integration for real-time verbal Q&A, reducing problem-solving delays during meetings.
- Built a scalable backend with WebRTC, tRPC, Drizzle ORM, NeonDB, and subscription billing via Polar API.

ChatAway - Real-time Chat App | Demo | Live Site | GitHub

- Built a real-time WebSocket-based chat application, addressing delays in teamwork by providing immediate messaging and media sharing.
- Implemented Socket.IO with OAuth (GCP), JWT auth, and global state management with Zustand.

Plant Disease Detection | Demo | GitHub

- Built a full-stack AI app to detect plant diseases (Early/Late Blight, Healthy) using a TensorFlow/Keras model with 95% accuracy, delivering real-time predictions with confidence scores via an integrated API.